

Vector WS 2

1. The straight line l_1 has the vector equation $\tilde{u} = 3\tilde{i} + \tilde{j} + 2\tilde{k} + \lambda(\tilde{i} - \tilde{j} + 4\tilde{k})$. The straight line l_2 has the vector equation $\tilde{u} = 4\tilde{j} - 2\tilde{k} + \mu(\tilde{i} - \tilde{j})$, where λ and μ are parameters. The lines l_1 and l_2 intersect at A and the acute angle between l_1 and l_2 is θ .

- What are the coordinates of A?
- Find the value of $\cos\theta$, giving answer as a simplified fraction.

2. The two lines $\hat{r} = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix}$ and $\hat{q} = \begin{pmatrix} -1 \\ 6 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ intersect at (a, b, c) .

- Find (a, b, c) .
- Show that $\hat{r}$ and $\hat{q}$ are perpendicular.
- Find a vector which is perpendicular to both $\hat{r}$ and $\hat{q}$.

3. a. Find the parametric equations of the line $\hat{l} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ passing through $A(2, -1, 3)$ which is parallel to $\hat{v} = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}$.

- Show that the point $B = (10, 5, -1)$ lies on this line.

4. A molecule of ammonia ion $[NH_4]^+$ is structured with four hydrogen atoms at the vertices of a regular tetrahedron and the nitrogen atom at the centroid. The bond angle is the angle formed by the $H - N - H$ combination. It is the angle between the lines that join the nitrogen atom to two of the hydrogen atoms.

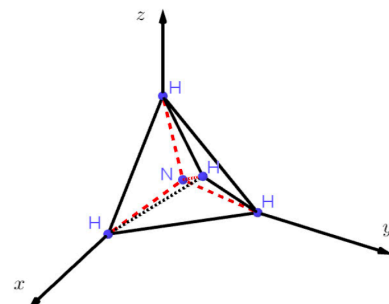
Take the vertices of the tetrahedron to be the points

$A(1, 0, 0)$, $B(0, 1, 0)$, $D(0, 0, 1)$ and $E(1, 1, 1)$ where A, B, D and E

representing the hydrogen atoms as shown in the figure. The nitrogen atom is represented by point N .

- Given that $\overrightarrow{NA} + \overrightarrow{NB} + \overrightarrow{ND} + \overrightarrow{NE} = 0$, show that the centroid is given by $N\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$.

- Show that the bond angle is approximately 109.5° .



5. a. Consider the sphere given by the Cartesian equation

$$x^2 + y^2 + z^2 + 2x - 4z - 4 = 0. \text{ Show that the vector equation of the sphere is } |\hat{r} - \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}| = 3|.$$

- Find the points of intersection between the sphere and the line $\hat{r} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$.

- Show that the line $\hat{r} = -\hat{i} + \hat{j} - \hat{k} + \mu\hat{j}$ is tangent to the sphere.

6. Find the equation of the sphere with the end points of a diameter $(2, 3, 5)$ and $(4, -3, 9)$.

7. The position vectors of the points P, Q and R are $8\tilde{i} + 4\tilde{j} - 3\tilde{k}$, $6\tilde{i} + 3\tilde{j} - 4\tilde{k}$ and $7\tilde{i} + 5\tilde{j} - 5\tilde{k}$ respectively. Find the angle between $\overrightarrow{PQ}$ and $\overrightarrow{QR}$.

8. For the curves whose vector equations are given, find the parametric equation and the Cartesian equation.

- $\vec{r} = \left(t + \frac{1}{t}\right)\tilde{i} + \left(t - \frac{1}{t}\right)\tilde{j}, t \neq 0$

- $\vec{r} = (u + 3)\tilde{i} + (u^2 - 5)\tilde{j}, u \in \mathbb{R}$

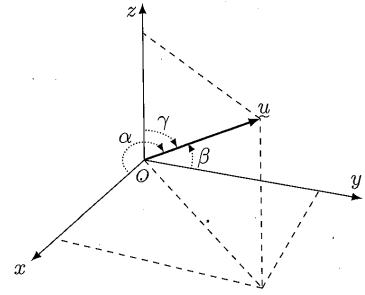
9. Graph the curve represented by each vector function.

- $\begin{pmatrix} 0 \\ t \\ t^2 \end{pmatrix}$ for $0 \leq t \leq 3$

- $\begin{pmatrix} \cos t \\ t \\ \sin t \end{pmatrix}$ for $0 \leq t \leq 2\pi$

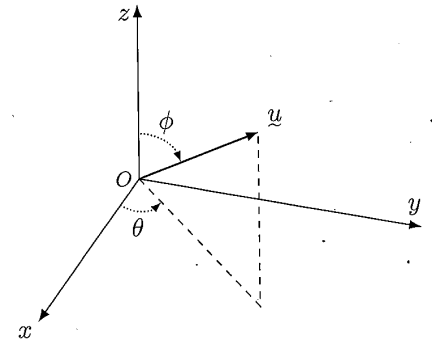
10. The direction cosines of a vector $\vec{u} = a\vec{i} + b\vec{j} + c\vec{k}$ are the cosines of the angles from the vector to each of the coordinate axes. Let α, β and γ be the angles of vector $\vec{u}$ from the x, y and z axes respectively.

- Show that $\cos \alpha = \frac{a}{|\vec{u}|}$
- Deduce that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$.



11. Let θ and ϕ be the angles shown in the diagram below. Let the direction cosines be defined as in the previous question. Show that the direction cosines of the vector $\vec{u}$ can be expressed as the following.

$$\begin{aligned}\cos \alpha &= \cos \phi \cos \theta \\ \cos \beta &= \cos \phi \sin \theta \\ \cos \gamma &= \sin \phi\end{aligned}$$



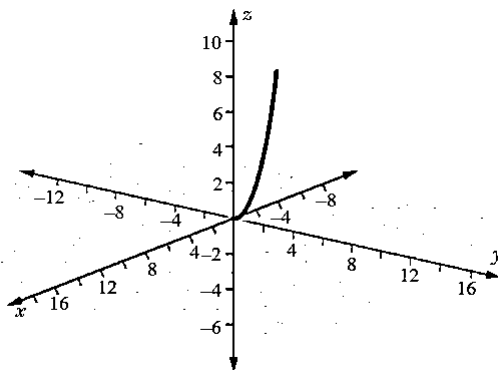
12. A particle moves according to the path defined parametrically by $\vec{r}(t) = (2t^3 + t)\vec{i} + (4 - t^2)\vec{j} + (3t + 1)\vec{k}$, where t is time in seconds.
- Write down an expression for the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.
 - Find the speed and acceleration of the particle after one second.

Answer

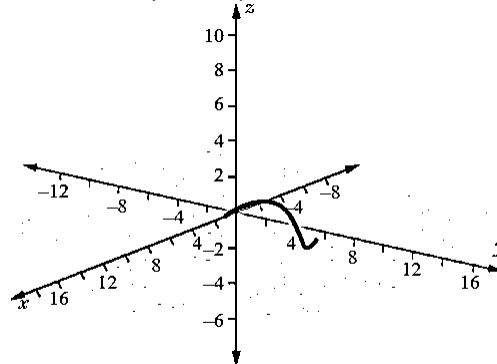
- a. $A: 2\vec{i} + 2\vec{j} - 2\vec{k}$; b. $\frac{1}{3}$; 3. $A = 9\sqrt{2}u^2$
- a. $(-3, 2, 1)$; c. $(-114, -60, 78)$ or $(19, 10, -13)$
- Proving
- Proving
- b. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$
- $(x - 3)^2 + y^2 + (z - 7)^2 = 14$
- 120°
- a. Parametric: $x = t + \frac{1}{t}, y = t - \frac{1}{t}, t \neq 0$, Cartesian: $x^2 - y^2 = 4, x + y \neq 0$; b. Parametric: $x = u + 3, y = u^2 - 5, u \in \mathbb{R}$, Cartesian: $y = (x - 3)^2 - 5$ or $y = x^2 - 6x + 4$

9.

a.



b. A helix (spiral) of radius 1 unit revolving around the y -axis, with endpoints $(-1, 0, 0)$ and $(-1, \pi, 0)$.



10. Proving

11. proving

12. a. $\vec{v}(t) = (6t^2 + 1)\vec{i} - (2t)\vec{j} + 3\vec{k}$, $\vec{a}(t) = (12t)\vec{i} - 2\vec{j}$; b. $|\vec{v}(1)| = \sqrt{62}, |\vec{a}(1)| = 2\sqrt{145}$